Insulation Resistance Tester



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Hand driven Megger; Megger

What it is used for

An insulation resistance tester applies a specified test voltage to assess insulation resistance in isolated electrical equipment or circuits. Use the correct test voltage and follow electrical isolation procedures before testing or reconnecting equipment.

Main parts

- Insulation resistance tester or hand-driven megger
- Test leads and insulated probes
- Positive and negative test terminals
- Test voltage selector or crank mechanism
- Display, scale, or resistance indicator
- Protective case and test accessories

Before you start

1. Confirm the circuit or equipment is suitable for insulation resistance testing.
2. Identify the circuit and isolate it from every power source.
3. Apply lockout/tagout procedures and verify absence of voltage with an appropriate tester.
4. Disconnect sensitive electronic devices, control modules, and components that could be damaged by test voltage.
5. Select the test voltage according to the equipment rating and applicable procedure.
6. Inspect the tester, leads, probes, and insulation for damage.

How to use

1. Keep the circuit isolated and ensure it cannot be energised during testing.
2. Connect the test leads to the designated tester terminals.
3. Connect the probes across the specified conductors or between conductor and earth as required.
4. Confirm that the selected test voltage is suitable for the equipment under test.
5. For a hand-driven model, operate the handle at the speed specified by the manufacturer.
6. For an electronic model, activate the test according to its instructions.
7. Maintain the test for the prescribed duration and observe the resistance reading.
8. Record the reading, test voltage, duration, equipment identification, and environmental conditions if required.
9. Release the test control and wait for the instrument to indicate completion.
10. Allow the tested circuit to discharge fully; use the tester's discharge function if provided.

1. Verify that residual voltage has dissipated before touching the conductors.
2. Disconnect the test leads and restore connections only after testing is complete and safe.
3. Remove isolation only under the authorised procedure and record the test result.

⚠ Safety

- Only qualified or authorised personnel should perform electrical insulation tests.
- Never apply test voltage to an energised circuit.
- Do not touch test leads or conductors during a test.
- Use the correct test voltage to avoid damaging insulation or connected components.
- Treat tested cables and capacitive equipment as charged until fully discharged.
- Do not test in wet conditions or use damaged leads and probes.

Care & storage

- Keep the tester and accessories clean, dry, and protected from impact.
- Inspect leads, probes, insulation, and terminals before each use.
- Check calibration or verification status according to the maintenance schedule.
- Replace damaged accessories with compatible approved parts.
- Store the tester in its protective case and record faults.

Common mistakes to avoid

- Testing a circuit without verifying electrical isolation.
- Selecting an unsuitable test voltage for the equipment.
- Touching conductors before the tested circuit has discharged.
- Failing to disconnect sensitive electronics before applying test voltage.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar